RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Chapter 8A: Graphing Techniques

SYLLABUS INCLUDES

- Equations of asymptotes, axes of symmetry, and restrictions on the possible values of x and/or y
- Graphical Interpretation of
  (i) $\mathrm{f}'(x) > 0$, $\mathrm{f}'(x) = 0$ and $\mathrm{f}'(x) < 0$
  (ii) $\mathrm{f}''(x) > 0$ and $\mathrm{f}''(x) < 0$
- Determining the nature of the stationary points (local maximum and minimum points and points of inflexion) analytically, in simple cases, using the first derivative test or the second derivative test
- Locating maximum and minimum points using a graphing calculator
- Important characteristics of graphs such as symmetry, intersections with the axes, turning points and asymptotes of the following:
  $y^2 = ax \ ; \ x^2 = by$
  $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$
  $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1 ; \ \dfrac{y^2}{b^2} - \dfrac{x^2}{a^2} = 1$
  $y = \dfrac{ax+b}{cx+d}$
  $y = \dfrac{ax^2+bx+c}{dx+e}$
- Use of a graphing calculator or a graphing software to graph a given function
- Simple parametric equations and their graphs

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1.2 Axes of symmetry
- 1.2.1 Symmetry about the *x*-axis or *y*-axis
- 1.2.2 Symmetry about other lines

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2.1 Graphs of functions of the form $\mathrm{f}(x)=\dfrac{ax+b}{cx+d}$, where $c \neq 0$, and the numerator and denominator have no common factor if $a \neq 0$

2.2 Graphs of functions of the form $\mathrm{f}(x)=\dfrac{ax^2+bx+c}{dx+e}$, where $a \neq 0$, $d \neq 0$ and the numerator and the denominator have no common factors

3 Conic Sections

3.1 Parabola

3.2 Circle

3.3 Ellipse

3.4 Hyperbola

4 Parametric Equations and their graphs

Appendix

I Sketching ellipse and hyperbola using a graphing calculator

1 CHARACTERISTICS OF GRAPHS

In sketching graphs, certain features of a curve can be observed, or easily calculated, from the equation of the curve. Some of the important features that we can consider are intersections with the axes, axes of symmetry, asymptotes, restrictions on the possible values of x and/or y turning points etc.

1.1 Intersections with the axes

It is usually easy to determine whether the curve $y = \mathrm{f}(x)$ meets the x-axis and/or y-axis. The x intercepts are found by setting $y = 0$ in the equation of the curve and solving for x. Similarly, the y intercepts are found by setting $x = 0$ in the equation of the curve and solving for y.

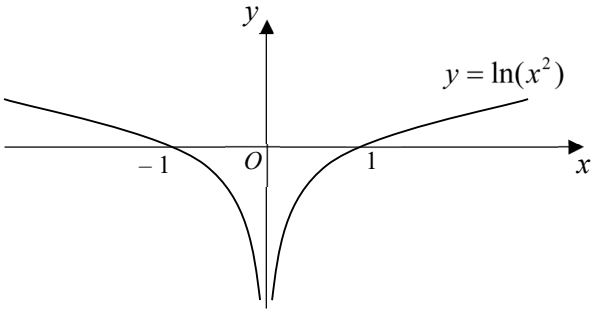
For example, the curve with equation $\frac{x^2}{4} - \frac{y^2}{9} = 1$ clearly intersects the x-axis at $(2,0)$ and $(-2,0)$ but does not cross the y-axis.

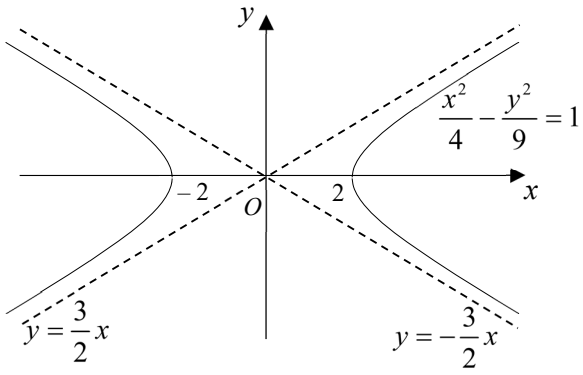
1.2 Axes of symmetry

1.2.1 Symmetry about the *x*-axis or *y*-axis

If the equation of the curve is unchanged when x is replaced by $(-x)$, the graph is symmetrical about the y-axis. Similarly, if the equation of the curve is unchanged when y is replaced by $(-y)$, the graph is symmetrical about the x-axis.

For example, the graph of the curve with equation $y = \ln(x^2)$ is symmetrical about the y-axis while the graph of the curve with equation $\frac{x^2}{4} - \frac{y^2}{9} = 1$ is symmetrical both about the x-axis and y-axis.





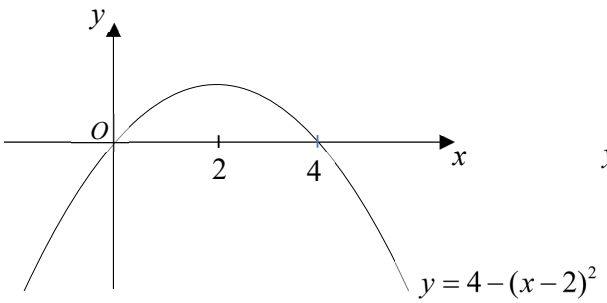
Note:

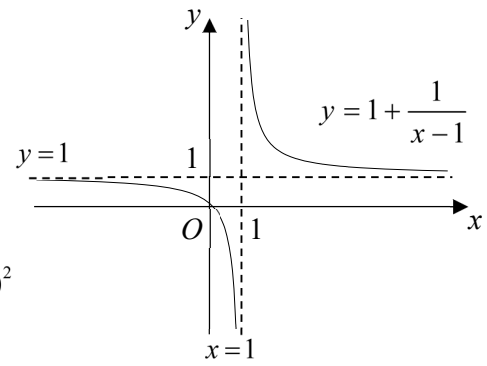
The dotted lines $y = \frac{3}{2}x$ and $y = -\frac{3}{2}x$ are asymptotes to the curve $\frac{x^2}{4} - \frac{y^2}{9} = 1$. In the case of the curve $y = \ln(x^2)$, the line $x = 0$ is an asymptote to the curve. We will learn more about asymptotes in section **1.4.**

1.2.2 Symmetry about other lines

Besides symmetry about the axes, the graph of the curve could possibly have other axes of symmetry.
For the curve with equation $y = 4 - (x-2)^2$, the graph is symmetrical about the line $x = 2$.

In the case of the curve with equation $y = 1 + \dfrac{1}{x-1}$, the lines of symmetry are $y = x$ and $y = -x + 2$.





Note:
To determine if a curve is symmetrical in the line $y = x$, we can easily do so by replacing both x with y and y with x. If the equation of the curve is unchanged, then the line $y = x$ is a line of symmetry.

However, it may not be easy to see from its equation that the curve with equation $y = 1 + \dfrac{1}{x-1}$ also has $y = -x + 2$ as a line of symmetry. We will be able to understand this better in Section 5 (Standard Transformation).

Exercise:
State a line of symmetry of the following curves whose equations are

(i) $y = \dfrac{8}{x^2 - x + 1}$, (ii) $y = \mathrm{e}^{-x^2}$. Answer: (i) $x = \dfrac{1}{2}$, (ii) $x = 0$

1.3 Restrictions on the possible values of *x* and/or *y*

It is sometimes useful to consider whether there are any restrictions on the possible values of *x* and/or *y* when we graph the curve $y = \mathrm{f}(x)$.

For instance, the curve with equation $y = \ln(2-x)$ is only well-defined when $x < 2$. Thus the graph of $y = \ln(2-x)$ lies only in the region where $x < 2$.

In the case of the curve with equation $y^2 = x - 1$, we observe that since $y^2 \geq 0$ for all $y \in \mathbb{R}$, then $x - 1 \geq 0 \Rightarrow x \geq 1$. Thus the graph lies only in the region where $x \geq 1$.

Example 1

It is given that that $y = \dfrac{x^2+3}{x+1}, \quad x \in \mathbb{R}, x \neq -1$. Without using a calculator, find the set of values that *y* can take.

Solution

If *y* can take the value *k*, the graphs of $y = \dfrac{x^2+3}{x+1}$ and $y = k$ must intersect.

At the point(s) of intersection,

$$k = \frac{x^2+3}{x+1}$$

$$kx + k = x^2 + 3$$

$$x^2 - kx + (3-k) = 0$$

For the quadratic equation to have real roots,

Discriminant ≥ 0

$$k^2 + 4k - 12 \geq 0$$

$$(k+6)(k-2) \geq 0$$

$$k \leq -6 \quad \text{or} \quad k \geq 2$$

Hence, set of values *y* can take is $(-\infty, -6] \cup [2, \infty)$.

Note:
Graphs of parabolas, circles, ellipses and hyperbolas (which will be dealt with in Section **3**) have certain restrictions on values of *x* and *y* as well.

Consider the graph of $x^2 + y^2 = 9$, which is a circle with centre at the origin and radius 3 units. We note that $-3 \leq x \leq 3$ and $-3 \leq y \leq 3$.

For the graph of hyperbola $9x^2 - 4y^2 = 36$, we observe that as $9x^2 = 36 + 4y^2$. Since $y^2 \geq 0$ for all $y \in \mathbb{R}$, we have $x^2 \geq 4$. Thus $x \leq -2 \text{ or } x \geq 2$.

1.4 Asymptotes

An asymptote of the curve with equation $y = \mathrm{f}(x)$ is a line or a curve that the graph of $y = \mathrm{f}(x)$ approaches when x or y is large. It gives an indication of the behaviour of the curve when $x \to \infty$ or $-\infty$ and/or when $y \to \infty$ or $-\infty$. In our syllabus, we will only concern ourselves with linear asymptotes.

Example 2

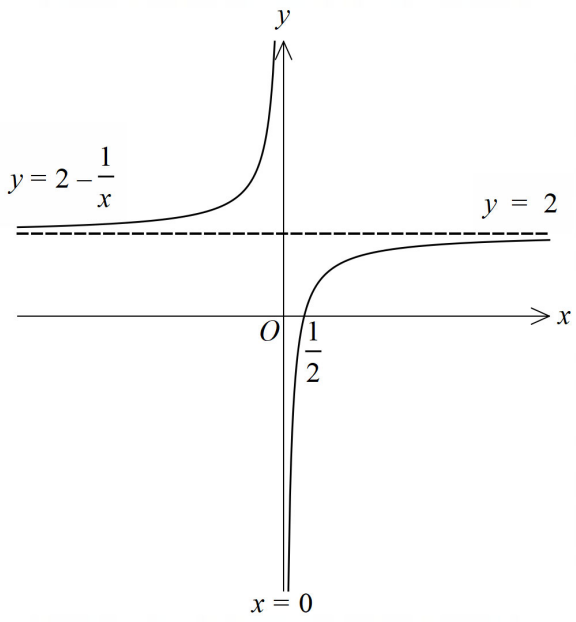
Find the equations of the asymptotes of the curve with equation $y = 2 - \frac{1}{x}$.

Solution

As $x \to \infty, y \to 2$ and hence $y = 2$ is a horizontal asymptote.
Note that, in fact, we also have as $x \to -\infty, y \to 2$.

As $x \to 0^-$ (i.e. approaching $x = 0$ from the left), $y \to \infty$. Thus $x = 0$ is a vertical asymptote.
Note also that as $x \to 0^+$ (i.e. approaching $x = 0$ from the right), $y \to -\infty$.

Graphically, we see that



Example 3

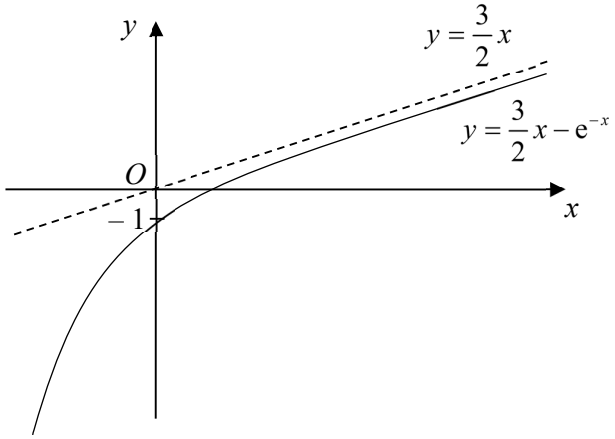
Find the equation(s) of the asymptote(s) of the curve with equation $y = \frac{3}{2}x - \mathrm{e}^{-x}$.

Solution

As $x \to \infty$, $\mathrm{e}^{-x} \to 0$ and hence $y \to \frac{3}{2}x$.

Thus $y = \frac{3}{2}x$ is an oblique asymptote.

Graphically, we have



Exercise:

State the equations of the asymptotes of the curve with equation $y = x - \frac{1}{x^2}$. Ans: $x = 0$, $y = x$

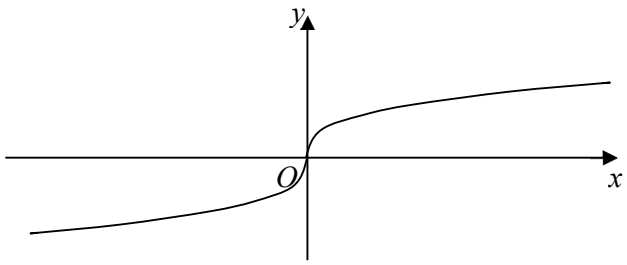
1.5 Shape of graphs

Having discussed intercepts and asymptotes of graphs, it remains for us to study the shape of the graph. In this section, we will discuss the concept of strictly increasing (decreasing), concavity as well as stationary points. These form the foundation to determine the shape of any given graph. Hence it is important for us to look at $\mathrm{f}'(x)$ to determine the shape of the graph.

1.5.1 MONOTONICITY & CONCAVITY

> A function f is said to be strictly **increasing** on an interval I if for all $b > a$ where $a, b \in I$, $\mathrm{f}(b) > \mathrm{f}(a)$.

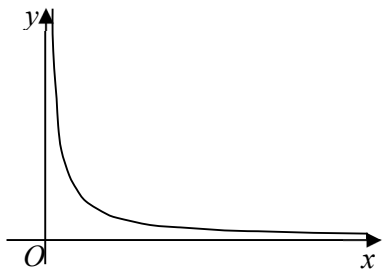
For example, $\mathrm{f} : x \mapsto x^{\frac{1}{3}}, \quad x \in \mathbb{R}$.



Similarly,

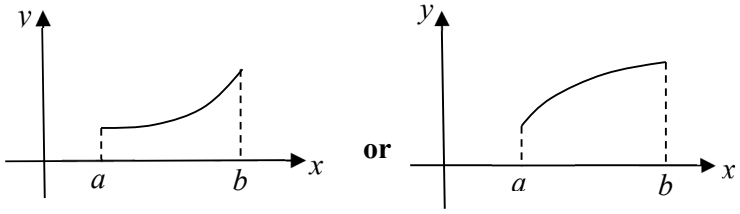
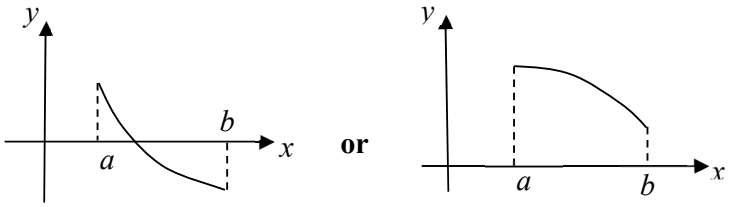
> A function f is said to be strictly **decreasing** on an interval I if for all $b > a$ where $a, b \in I$, $\mathrm{f}(b) < \mathrm{f}(a)$.

For example, $\mathrm{g} : x \mapsto \dfrac{1}{x}, \quad x \in \mathbb{R}, x > 0$.



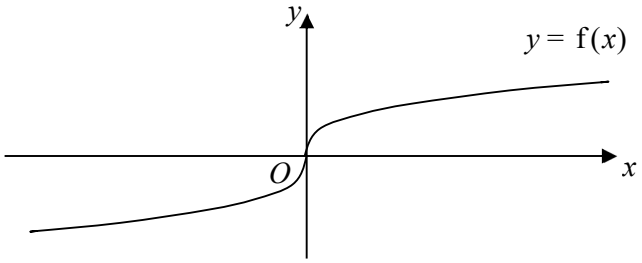
Consider a function f given by $y = \mathrm{f}(x),\ x \in I$.

- $\mathrm{f}'(x)$ denotes the rate of change of $\mathrm{f}(x)$ with respect to x, and f' is known as the first derivative or gradient function.
- $\mathrm{f}''(x)$ denotes the rate of change of gradient $\mathrm{f}'(x)$ with respect to x, and f'' is known as the second derivative.
- The sign of the first derivative of the function gives us information about its monotonicity (i.e. increasing or decreasing) while the sign of the second derivative of the function gives us information about its concavity (i.e. concave up or down).

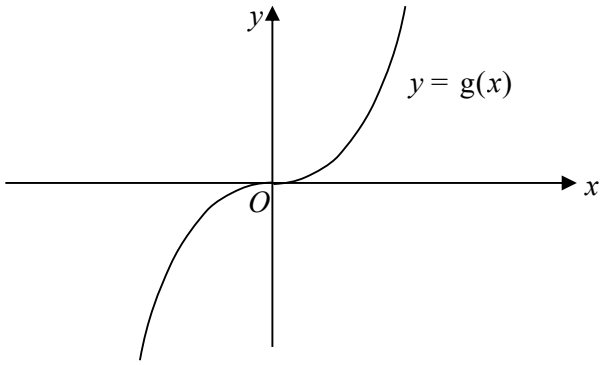
Sign of $\mathrm{f}'(x)$ over (a,b)	**Graphical Perspective**
If $\mathrm{f}'(x) > 0$, then f is strictly increasing over the interval (a,b).	
If $\mathrm{f}'(x) < 0$, then f is strictly decreasing over the interval x.	

Note:
The converse of the above statements are not true.
That is, a strictly increasing function can have points where the derivative $\mathrm{f}'(x)$ is 0 or undefined.

Consider the function $\mathrm{f}(x) = x^{\frac{1}{3}},\quad x \in \mathbb{R}$. The function is strictly increasing but $\mathrm{f}'(x) = \dfrac{1}{3x^{\frac{2}{3}}}$ is undefined at $x = 0$.



Consider the function $\mathrm{g}(x) = x^3, \quad x \in \mathbb{R}$. The function is strictly increasing but $\mathrm{g}'(0) = 0$.



Example 4

Prove that the function f defined by $\mathrm{f}(x) = x^3 - 6x^2 + 18x + 5, \ x \in \mathbb{R}$ is strictly increasing.

Solution

$\mathrm{f}(x) = x^3 - 6x^2 + 18x + 5, \ x \in \mathbb{R}$

Since $\mathrm{f}'(x) = 3x^2 - 12x + 18$

$= 3(x^2 - 4x) + 18$

$= 3[(x-2)^2 - 4] + 18$

$= 3(x-2)^2 + 6 > 0$ since $(x-2)^2 \geq 0 \quad \forall x \in \mathbb{R}$

The function f is strictly increasing. (proven)

Example 5

The function f defined by $\mathrm{f}(x) = 2\left(x - \frac{1}{2}\right) - \ln x$, where $x \in \mathbb{R}, \ x > 0$. Find $\mathrm{f}'(x)$ and show that the function is strictly increasing for $x > \frac{1}{2}$. Hence show that for $x > \frac{1}{2}$, $2\left(x - \frac{1}{2}\right) - \ln x > \ln 2$.

Solution

$\mathrm{f}'(x) = 2 - \frac{1}{x}$

When $x > \frac{1}{2}, \quad 0 < \frac{1}{x} < 2$ and hence $\frac{\mathrm{d}y}{\mathrm{d}x} = 2 - \frac{1}{x} > 0$.

Thus the curve is strictly increasing for $x > \frac{1}{2}$. (shown)

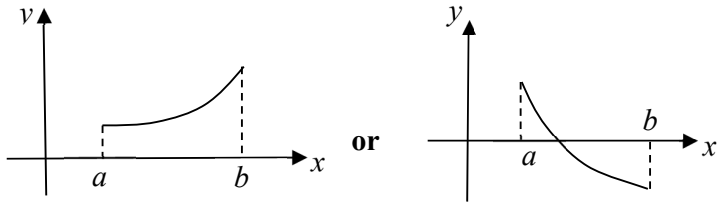
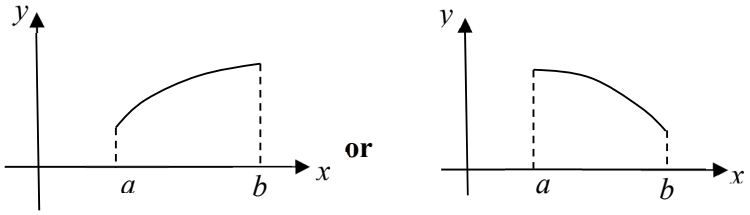
When $x = \frac{1}{2}, \quad \mathrm{f}\left(\frac{1}{2}\right) = 2\left(\frac{1}{2} - \frac{1}{2}\right) - \ln\frac{1}{2} = \ln 2$.

Since the function is strictly increasing for $x > \frac{1}{2}$, $\mathrm{f}(x) > \mathrm{f}\left(\frac{1}{2}\right)$

$2\left(x - \frac{1}{2}\right) - \ln x > \ln 2.$ (shown)

A function f is said to **concave up** on an interval I if for any $a, b \in I$, the line segment joining $(a, \mathrm{f}(a))$ and $(b, \mathrm{f}(b))$ lies above the graph of the function.

Similarly, a function f is said to **concave down** on an interval I if for any $a, b \in I$, the line segment joining $(a, \mathrm{f}(a))$ and $(b, \mathrm{f}(b))$ lies below the graph of the function.

Sign of $\mathrm{f}''(x)$ over (a,b)	**Graphical Perspective**
If $\mathrm{f}''(x) > 0$, then f is concave up over the interval x.	
If $\mathrm{f}''(x) < 0$, then f is concave down over the interval x.	

Note:

- $\mathrm{f}'(x) > 0$ means that as x increases, $\mathrm{f}(x)$ increases

  $\mathrm{f}'(x) < 0$ means that as x increases, $\mathrm{f}(x)$ decreases

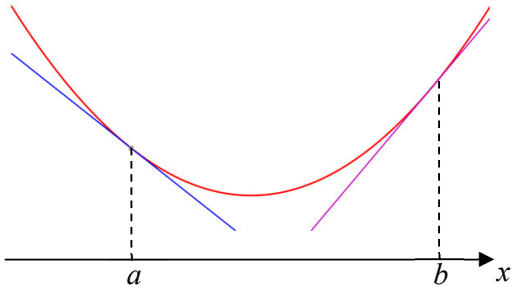
- $\mathrm{f}''(x) > 0$ means that as x increases, gradient $\mathrm{f}'(x)$ increases

  $\mathrm{f}''(x) < 0$ means that as x increases, gradient $\mathrm{f}'(x)$ decreases

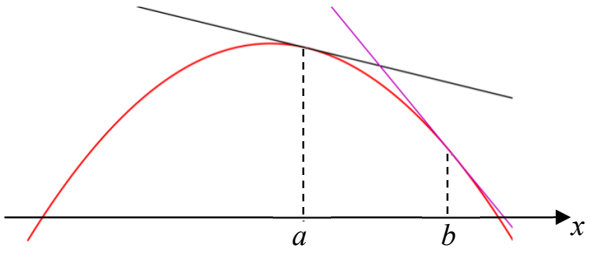
Remark:

We observe from the above that if a function f is **concave up** over an interval I, then the graph of the function lies *above* the tangent to the curve $y = \mathrm{f}(x)$ at $x = k$ for all $k \in I$.

Observe from the graph, for all $a, b \in I$ with $a < b$, $\mathrm{f}'(a) < \mathrm{f}'(b)$, i.e. gradient $\mathrm{f}'(x)$ increases.

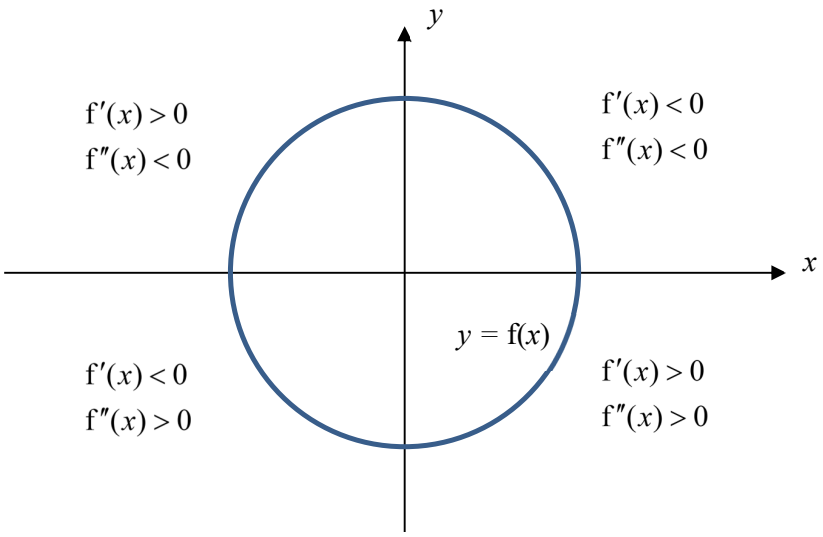


In the case where a function f is **concave down** over an interval I, then the graph of the function lies *below* the tangent to the curve $y = \mathrm{f}(x)$ at $x = k$ where $k \in I$.



Observe from the graph, for all $a, b \in I$ with $a < b$, $\mathrm{f}'(a) > \mathrm{f}'(b)$, i.e. gradient $\mathrm{f}'(x)$ decreases.

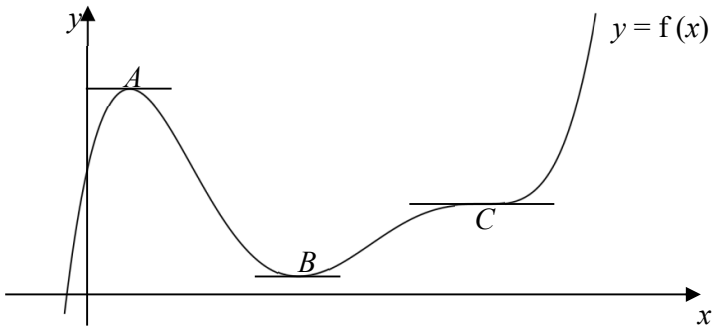
An example to illustrate concavity and sign of gradient:



1.5.2 STATIONARY POINTS, TURNING POINTS AND POINTS OF INFLEXION

If a curve with equation $y = \mathrm{f}(x)$ has a **stationary point** at $x = k$, then $\mathrm{f}'(k) = 0$.
i.e. at $x = k$, the gradient of the curve is zero and the tangent to the curve is parallel to the x-axis.

Let us now consider the following graph.
The stationary points of the graph are points A, B and C.



As we move along the curve from the slight left of *A* to the slight right of *A*, the gradient of the curve $\mathrm{f}'(x)$ changes from positive to zero and then negative. i.e. the curve $y = \mathrm{f}(x)$ increases and then decreases.
A is known as a *maximum turning point*.

As we move along the curve from the slight left of *B* to the slight right of *B*, the gradient of the curve $\mathrm{f}'(x)$ changes from negative to zero and then positive. i.e. the curve $y = \mathrm{f}(x)$ decreases and then increases.
B is known as a *minimum turning point*.

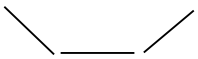
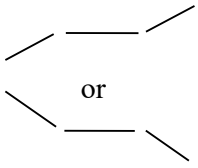
However, although the gradient is zero at *C*, the gradient of the curve does not change sign as we move along the curve from the slight left of *C* to the slight right of *C*.
C is known as a *stationary point of inflexion*.

Definition:
A **turning point** is a stationary point at which the first derivative of a function (i.e. gradient function) changes sign.

Thus we have the following methods for determining the nature of the stationary point.

Method 1 : First Derivative Test

Check the signs of $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ for $x = k^-$ and $x = k^+$.

x	k^- k k^+	k^- k k^+	k^- k k^+
$\dfrac{\mathrm{d}y}{\mathrm{d}x}$	+ve 0 − ve	−ve 0 + ve	+ve 0 + ve or −ve 0 − ve
Tangent			
Nature of Stationary Point	Maximum turning point	Minimum turning point	Stationary point of inflexion

Method 2 : Second Derivative Test

Check the sign of $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ at $x = k$.

If $\left.\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=k} < 0$, then $\left(k,\ \mathrm{f}(k)\right)$ is a maximum turning point.

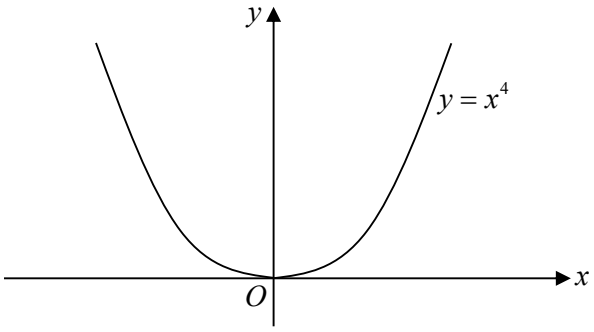
If $\left.\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=k} > 0$, then $\left(k,\ \mathrm{f}(k)\right)$ is a minimum turning point.

Note:

The converse of the 2 statements above is not true, i.e. $(k,\ \mathrm{f}(k))$ is a maximum turning point does not imply $\left.\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=k} < 0$ or $(k,\ \mathrm{f}(k))$ is a minimum turning point does not imply $\left.\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=k} > 0$.

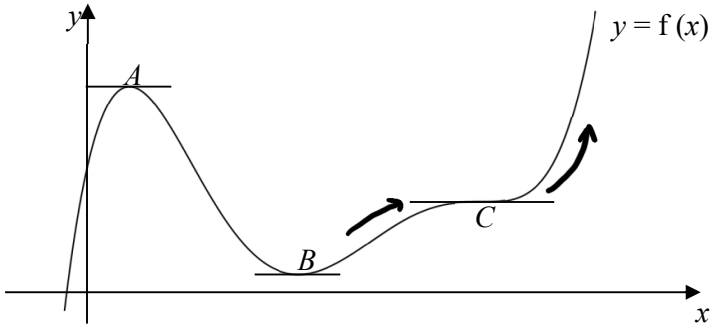
A counter-example would be the function $\mathrm{f}(x) = x^4,\ x \in \mathbb{R}$.

$(0,\ 0)$ is a minimum turning point but $\left.\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0} = 0$.

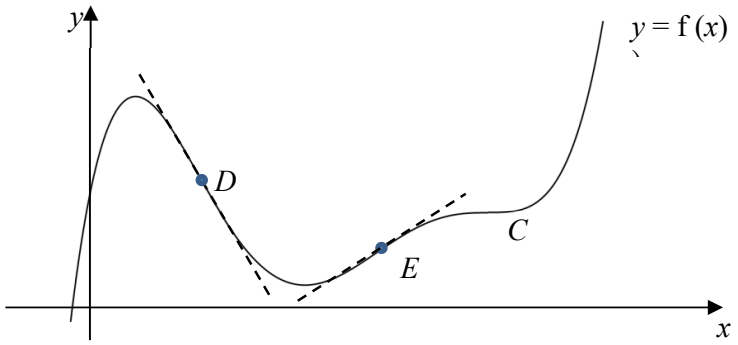


Thus there is no conclusion about the nature of the stationary point when $\left.\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=k} = 0$. In this case, the first derivative test must be used.

Let us look more closely at points of inflexion. A **point of inflexion** is a point on the curve at which the curve crosses its tangent and the concavity of the curve changes from up to down or vice versa.



Remark: *[Not in syllabus, but good to know.]*
In the above example, in addition to the point *C*, the points *D* and *E* are also points of inflexion. We call *D* and *E non-stationary points of inflexion* as the gradient at these points are not zero.



Example 6

Given $y = 2x^5 - 5x^3$, find $\frac{\mathrm{d}y}{\mathrm{d}x}$ and the coordinates of the stationary points. Determine the nature of each stationary point using the first derivative test. Sketch the graph of $y = 2x^5 - 5x^3$, giving the coordinates of all axial intercepts and stationary points.

Solution

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 10x^4 - 15x^2 = \quad 10x^2\left(x^2 - \frac{3}{2}\right) = 10x^2\left(x - \sqrt{\frac{3}{2}}\right)\left(x + \sqrt{\frac{3}{2}}\right)$$

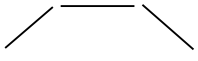
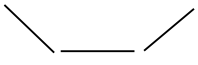
At stationary points, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \Rightarrow 10x^2\left(x - \sqrt{\frac{3}{2}}\right)\left(x + \sqrt{\frac{3}{2}}\right) = 0$

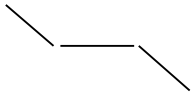
$$\Rightarrow x = 0 \quad \text{or} \quad x = \sqrt{\frac{3}{2}} \quad \text{or} \quad x = -\sqrt{\frac{3}{2}}$$

When $x = 0,\ y = 0$; when $x = -\sqrt{\frac{3}{2}},\ y = 3\sqrt{\frac{3}{2}}$; when $x = \sqrt{\frac{3}{2}},\ y = -3\sqrt{\frac{3}{2}}$

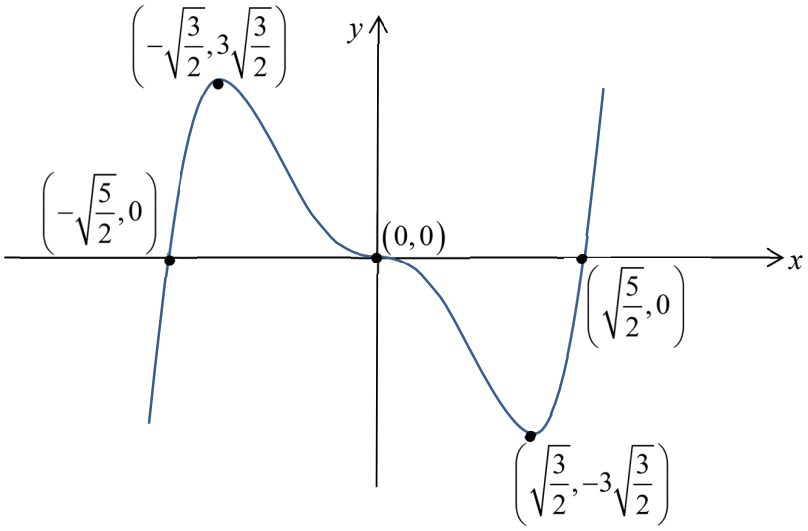
Coordinates of stationary points are $(0,0)$, $\left(\sqrt{\frac{3}{2}}, -3\sqrt{\frac{3}{2}}\right)$ and $\left(-\sqrt{\frac{3}{2}}, 3\sqrt{\frac{3}{2}}\right)$.

To check nature of stationary points,

x	$\left(-\sqrt{\frac{3}{2}}\right)^-$ $-\sqrt{\frac{3}{2}}$ $\left(-\sqrt{\frac{3}{2}}\right)^+$	$\left(\sqrt{\frac{3}{2}}\right)^-$ $\sqrt{\frac{3}{2}}$ $\left(\sqrt{\frac{3}{2}}\right)^+$
$\left(x - \sqrt{\frac{3}{2}}\right)\left(x + \sqrt{\frac{3}{2}}\right)$	+ve 0 − ve	−ve 0 + ve
$\frac{\mathrm{d}y}{\mathrm{d}x}$	+ve 0 − ve	−ve 0 + ve
Tangent		
Nature of Stationary Point	Maximum turning point	Minimum turning point

x	0^-	0	0^+
$x^2\left(x^2-\frac{3}{2}\right)$	$-$ve	0	$-$ve
$\frac{dy}{dx}$	$-$ve	0	$-$ve
Tangent			
Nature of Stationary Point	Stationary point of inflexion		

You may use the second derivative test if you wish.



NOTE:
If the question does not require the answers to be written in exact values, you may locate the maximum and minimum points using a graphing calculator.

SUMMARY FOR CURVE SKETCHING

A simple way to start any curve sketching question is to remember the following mnemonic: **S** (Shape/Stationary Points) **A** (Asymptotes) **I** (Intercepts) **L** (Labelling your graphs). In practice, we usually start from determining the asymptotes (if any), then the intercepts, followed by the shape, which includes the stationary points.

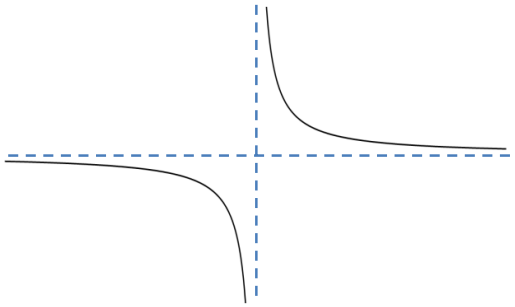
S	A	I	L
(**S**hape/**S**tationary Points)	(**A**symptotes)	(**I**ntercepts)	(**L**abel your graphs)

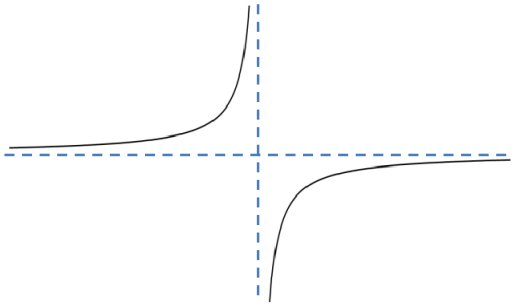
In the next section, we will learn how to sketch some standard curves such as graphs of rational functions and conic sections.

2 RATIONAL FUNCTIONS

2.1 Graphs of functions of the form $\mathrm{f}(x)=\dfrac{ax+b}{cx+d}$, where $c \neq 0$, and the numerator and denominator have no common factor if $a \neq 0$

In general, the graph of $y=\mathrm{f}(x)$ where $\mathrm{f}(x)=\dfrac{ax+b}{cx+d}$ where $c \neq 0$ takes one of the following





The dotted lines are known as **asymptotes**. They are lines which the curve $y=\mathrm{f}(x)$ approaches when x or y is large.

Note:

The graph of $y=\dfrac{ax+b}{cx+d}$ has no turning points. To understand this, we consider $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ which can be easily shown to be given by $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{ad-bc}{(cx+d)^2}$.

When $ad-bc \neq 0$, $\dfrac{\mathrm{d}y}{\mathrm{d}x} \neq 0$. Hence the graph of $y=\mathrm{f}(x)$ has no turning points.

When $ad-bc=0$, $\dfrac{\mathrm{d}y}{\mathrm{d}x}=0$ for all real values of x (except for $x=-\dfrac{d}{c}$). Hence $y=\mathrm{f}(x)$ will be a horizontal line (with the point $x=-\dfrac{d}{c}$ removed).

Example 7

Sketch the graph of $y=\dfrac{x-1}{x+2},\ x\neq -2$, labelling clearly any intersections with the axes, turning points and asymptotes.

Solution

S The function $y=\dfrac{x-1}{x+2}$ is of the form $y=\dfrac{ax+b}{cx+d}$, thus the graph of $y=\dfrac{x-1}{x+2}$ has no turning (stationary) points.

A *To find the vertical asymptote:*

We observe that as $x\to -2^-$, $y\to +\infty$. Also, we have as $x\to -2^+$, $y\to -\infty$.

$\therefore$ The line $x=-2$ is a vertical asymptote for the graph of $y=\dfrac{x-1}{x+2}$.

Simply put, vertical asymptotes for rational functions occur at points where the denominator is zero, provided the numerator and denominator have no common factors.

To find the horizontal asymptote:

We need to consider what happens to y as $x\to \pm\infty$.

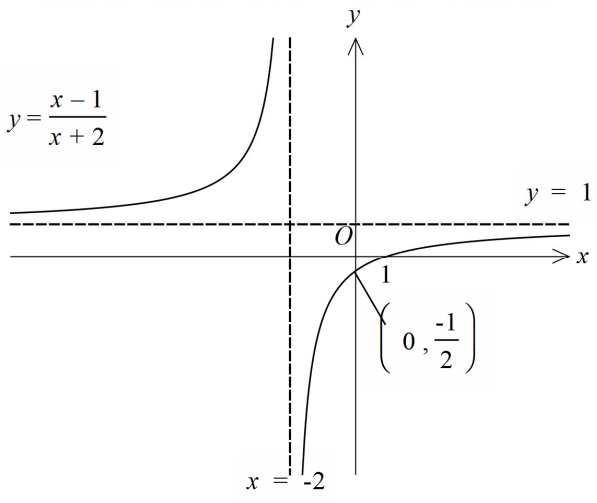
Express $y=\dfrac{x-1}{x+2}$ in the form $y=1-\dfrac{3}{x+2}$.

As $x\to \pm\infty$, $y\to 1$ (as $\dfrac{3}{x+2}\to 0$), i.e. the graph "approaches" the line $y=1$ as $x\to \pm\infty$.

$\therefore$ The line $y=1$ is a horizontal asymptote for the graph of $y=\dfrac{x-1}{x+2}$.

I When $x=0,\quad y=-\dfrac{1}{2}$ and when $y=0,\quad x=1.$

L

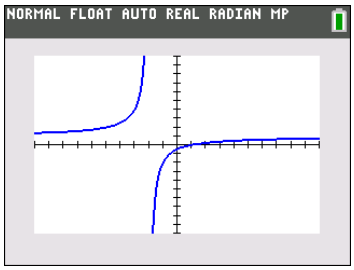
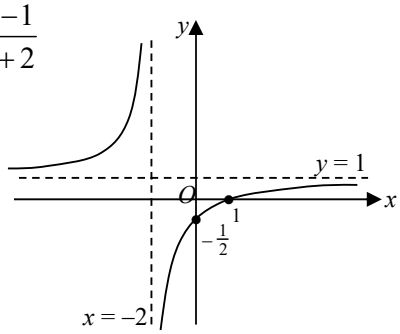


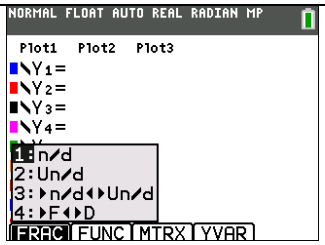
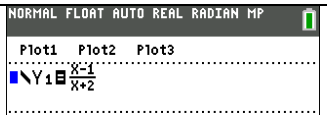
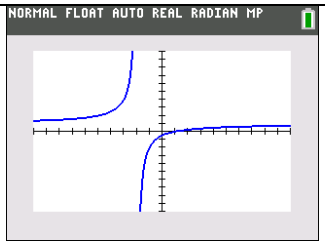
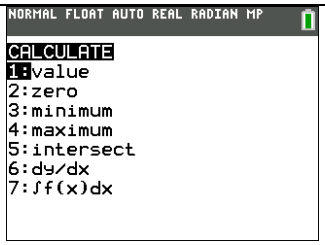
Note :

The GC gives the shape of the graph. In many cases, you may need to resize the window to capture the entire graph, or to change window settings to capture critical features of the graph.

We can use the GC to find intersections with axes and turning points, but **it does not draw asymptotes**! To determine if a graph has any asymptotes, you need to examine its equation.

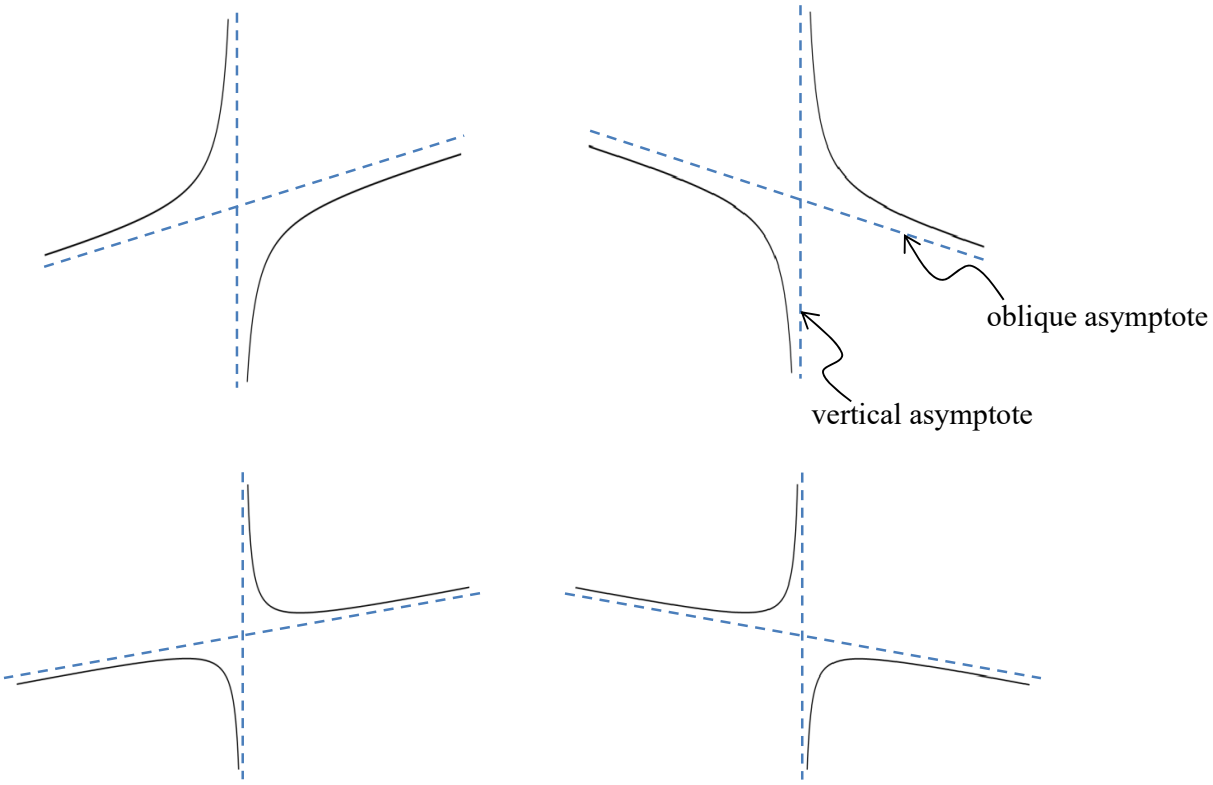
To illustrate this point, we shall investigate a case by sketching the graph of $y = \dfrac{x-1}{x+2}$.

What you see	What it should be
$y = \dfrac{x-1}{x+2}$  **Remarks** : There are 2 asymptotes, but both are not shown.	$y = \dfrac{x-1}{x+2}$ 

GC Keystrokes	*Screenshots*
1. Press alpha followed by y= 2. Select **1: n / d**	
3. Key in the numerator and denominator	
4. Press zoom 5. Select **6: ZStandard** (standard window of $-10 \le x \le 10,\ -10 \le y \le 10$)	
6. Press 2nd trace to open [calc] page 7. Select **1: value** (to calculate y-intercept) **2: zero** (to calculate x-intercept) **3: minimum** (to calculate min. turning pt.) **4: maximum** (to calculate max. turning pt.)	

2.2 Graphs of functions of the form $\mathrm{f}(x)=\dfrac{ax^2+bx+c}{dx+e}$, where $a\neq 0$, $d\neq 0$ and the numerator and the denominator have no common factors

In general, the graph of $y=\mathrm{f}(x)$ where $\mathrm{f}(x)=\dfrac{ax^2+bx+c}{dx+e}$ where $a\neq 0$, $d\neq 0$ takes one of the following forms.



The dotted lines are the **asymptotes**. There is a vertical asymptote and an oblique asymptote. The graph of $y=\mathrm{f}(x)$ may or may not have turning points depending on the values of *a*, *b*, *c*, *d* and *e*. We will now look at an example to illustrate how to sketch graphs of this form.

Special Features :

- Graph appears in opposite "quadrants".
- Graph has 1 vertical asymptote and 1 oblique asymptote.
- Case 1 : When the graph has no turning points, then the range of the function is $\mathbb{R}$.
  Case 2 : When the graph has 2 turning points (with (a,b) as minimum point, and (m,n) as maximum point), then the range of the function is $(-\infty,n]\cup[b,\infty)$.

Example 8

Sketch the graph of $y=\dfrac{x^2-3x-4}{x-1}$, $x\neq 1$, labelling clearly any intersections with the axes, turning points and asymptotes.

Solution

S $y=\dfrac{x^2-3x-4}{x-1}=x-2-\dfrac{6}{x-1} \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x}=1+\dfrac{6}{(x-1)^2}\neq 0$.

Hence, there are no stationary points on the curve.

A *To find the vertical asymptote:*

As $x\to 1^-$, $y\to +\infty$ and as $x\to 1^+$, $y\to -\infty$

Hence the graph has a vertical asymptote $x=1$.

As $x\to\infty$, y does not approach a constant, so there is no horizontal asymptote. However, we have an oblique asymptote.

To find the oblique asymptote :

Consider what happens to y as $x\to\pm\infty$.

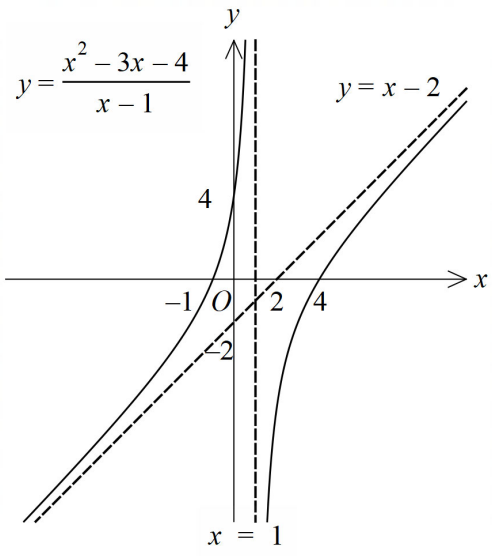
We first express $y=\dfrac{x^2-3x-4}{x-1}$ in the form $y=x-2-\dfrac{6}{x-1}$.

As $x\to\pm\infty$, $y\to x-2$ (since $\dfrac{6}{x-1}\to 0$), i.e. the graph "approaches" the line $y=x-2$ as $x\to\pm\infty$.

$\therefore$ The line $y=x-2$ is an oblique asymptote for the graph of $y=\dfrac{x^2-3x-4}{x-1}$.

I When $x=0,\ y=4$ and when $y=0,\ x=-1,\ 4$.

L

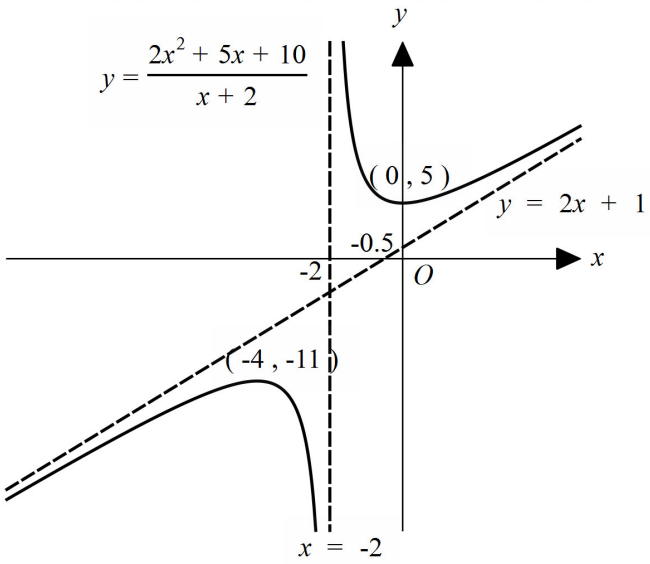


Example 9

Sketch the graph of $y = \dfrac{2x^2 + 5x + 10}{x + 2}$, $x \neq -2$, labelling clearly any intersections with the axes, turning points and asymptotes.

Solution

$$y = \frac{2x^2 + 5x + 10}{x + 2} = 2x + 1 + \frac{8}{x + 2}$$



Note:
In general, if the equation of the curve is known, we can use the GC to find the shape, intersections with axes and turning points of the graph. However, we need to examine the equation to determine the asymptotes.

Example 10

The curve C has equation $y = x + k + \dfrac{k+2}{x-k}$, where k is a constant, $k \neq -2$. Find the range of values of k for which the curve C cuts the x-axis at two distinct points.

It is given that $k = 2$.

(i) Sketch the curve C, stating its asymptotes, stationary points and axial intercepts if any.

(ii) The two asymptotes of C intersect at point P. Show that P lies on the line $y = mx + (4 - 2m)$ for all real values of m.

Hence, state the range of values of m for which the line $y = mx + (4 - 2m)$ does not cut the curve C.

Solution

$x+k+\dfrac{k+2}{x-k}=0$, $k\neq -2$

$x^2-k^2+k+2=0$

Since the curve C cuts the x-axis at two distinct points, the discriminant is positive.

$-4\left(-k^2+k+2\right)>0$

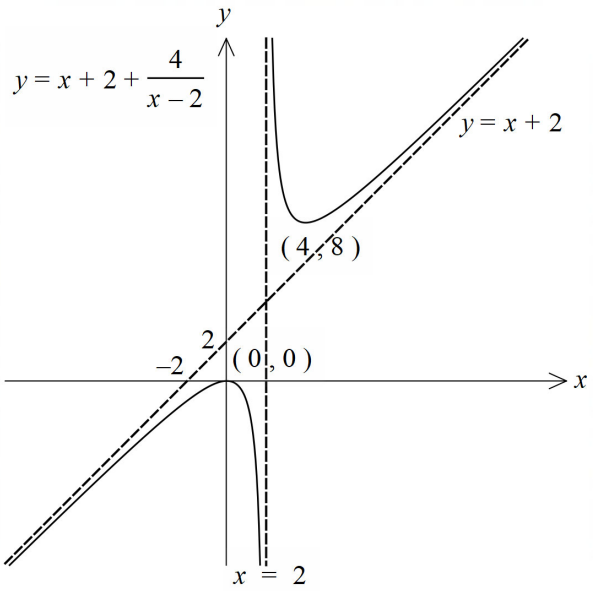
$k^2-k-2>0$

$(k+1)(k-2)>0$

$k<-1$ or $k>2$

Since $k\neq -2$, the range of values of k are $(-\infty,-2)\cup(-2,-1)\cup(2,\infty)$.

(i)



(ii) $y=x+2$ ------(1) $x=2$ ----------(2) $\Rightarrow y=4$

$\therefore$ $P(2,4)$

Any line that passes through $P(2,4)$ satisfies the following equation,

$y-4=t(x-2),\quad t\in\mathbb{R}$

$y=4+tx-2t$

$y=tx+(4-2t)$, where $t=m$.

$\therefore$ P lies on the line $y=mx+(4-2m)$ for all real values of m. (shown)

From the graph, if the line does not intersect C, $m\leq 1$.

NOTE : Regardless of the value of *m*, the line $y = mx + (4 - 2m)$ will pass through the point $(2, 4)$. The value of *m* represents the gradient of the line. When *m* >1, the line will still pass through the point $(2, 4)$ and has gradient more than 1 (that is, steeper than the oblique asymptote). This will result in intersecting the graph twice.

In summary,

$y = \dfrac{ax+b}{cx+d}$, where $c \neq 0$, numerator and denominator have no common factor	Express $y = \dfrac{ax+b}{cx+d}$ in the form $y = p + \dfrac{q}{cx+d}$, where *p* and *q* are constants. • Vertical asymptote : $x = -\dfrac{d}{c}$ • Horizontal asymptote : $y = p = \dfrac{a}{c}$ If $a = 0$, the horizontal asymptote is $y = 0$, i.e. x-axis
$y = \dfrac{ax^2+bx+c}{dx+e}$ where $a \neq 0,\ d \neq 0$, numerator and denominator have no common factor	Express $y = \dfrac{ax^2+bx+c}{dx+e}$ in the form $y = (px+q) + \dfrac{r}{dx+e}$, where $p,\ q$ and *r* are constants, $p \neq 0$. • Vertical asymptote : $x = -\dfrac{e}{d}$ • Oblique asymptote : $y = px + q$

Note:
In general when we deal with graphs of rational functions, we can apply similar strategies discussed earlier to determine the equations of the asymptotes. That is,

- vertical asymptotes occur at values of *x* where the denominator is zero, provided the numerator and denominator have no common factors.
- horizontal or oblique asymptotes can be determined by writing the rational expression in a form of a polynomial plus a proper fraction.

Example 11

Determine the equations of any asymptotes of the following curves with equations

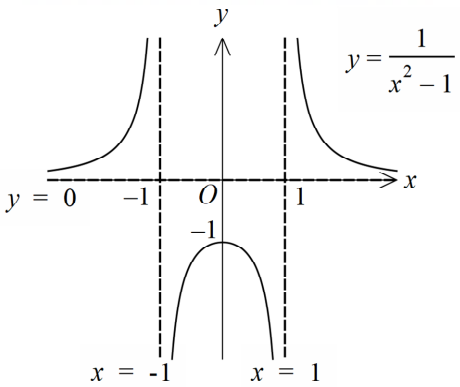
(a) $y=\dfrac{1}{x^2-1}$, **(b)** $y=\dfrac{x(2x+3)}{x^2-1}$ and sketch the curves on separate diagrams.

Remarks : *While **(a)** and **(b)** may not be the type of rational functions specifically mentioned in the syllabus, we are still expected to know how to sketch the graph of these functions. In such cases, proper graphing techniques (S-A-I-L mentioned in page 16) and graphing calculator skills become very relevant and important.*

Solution

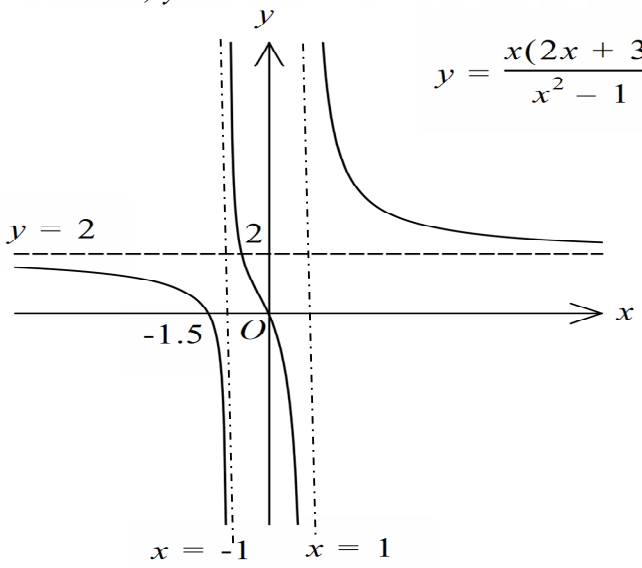
(a) $y=\dfrac{1}{x^2-1}$

Asymptotes are $x=\pm 1$, $y=0$.



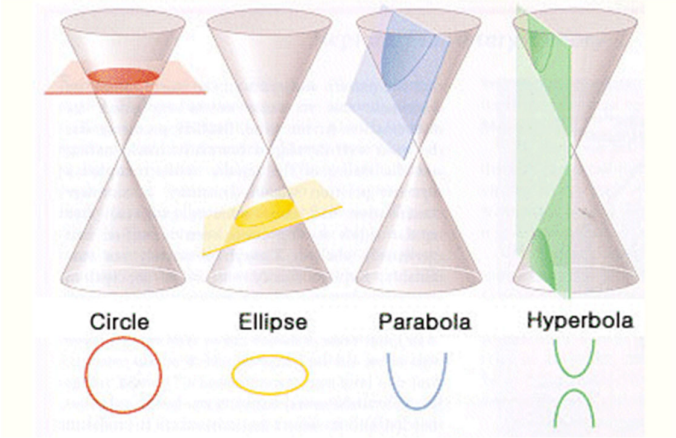
(b) $y=\dfrac{x(2x+3)}{x^2-1}=\dfrac{2x^2+3x}{x^2-1}=\dfrac{2(x^2-1)+2+3x}{x^2-1}=2+\dfrac{3x+2}{x^2-1}$

Asymptotes are $x=\pm 1$, $y=2$



3 CONIC SECTIONS

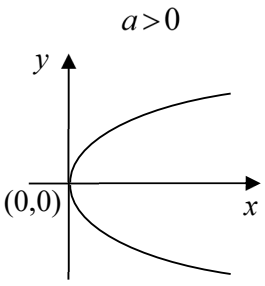
Conics are curves that can be obtained from the intersections of a cone and a plane. In this section, we will look at the equations of circle, ellipse, and hyperbola and some of their characteristics.

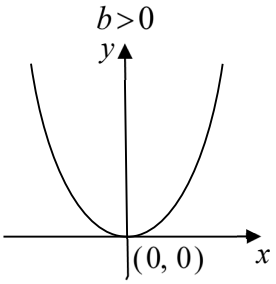


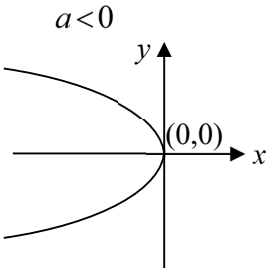
3.1 Parabola

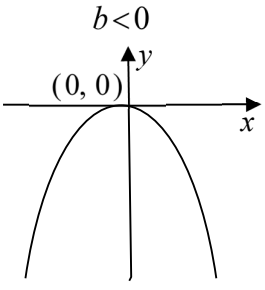
A parabola is a symmetrical open plane curve formed by the intersection of a cone with a plane parallel to its side. The path of a projectile under the influence of gravity follows a curve of this shape.

The equation of parabola with vertex (0,0) is given by
$y^2 = ax$ or $x^2 = by$









This graph is symmetrical about x-axis.

This graph is symmetrical about y-axis.

Note:
When sketching a parabola, show clearly its symmetry about $y=0$ or $x=0$. It is also necessary to label the vertex clearly.

Example 12
Sketch, on separate diagrams, the parabolas given by the equations

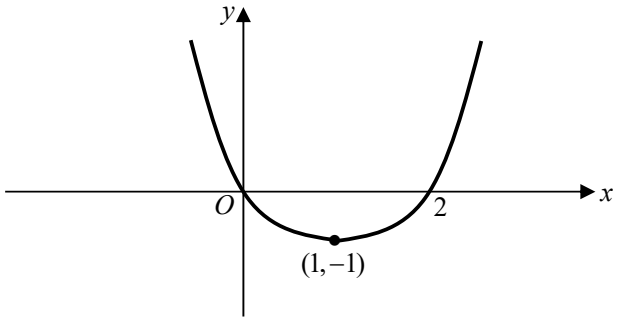
(i) $y=x^2-2x$

(ii) $(y-1)^2=2x+2,$

(iii) $2y^2-4x+10=0$.

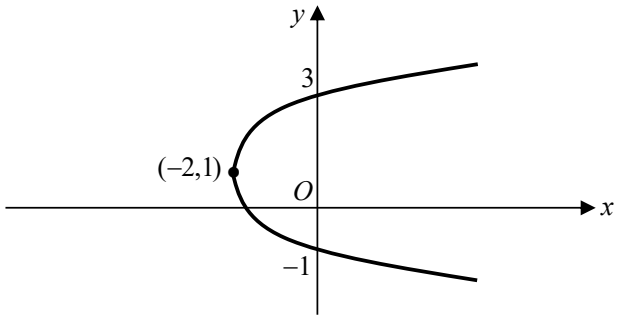
Solution

(i) $y=x^2-2x \Rightarrow y=(x-1)^2-1 \Rightarrow (x-1)^2=(y+1)$
Vertex of parabola is $(1,-1)$. Note that $b = 1 > 0$.
When $y=0$, $x=0$ or 2.



Of course, you can also continue to approach this question in the more familiar manner.
$y=x^2-2x \Rightarrow y=x(x-2) \Rightarrow x=0,2$ are x-intercepts.
Minimum point is $(1,-1)$.

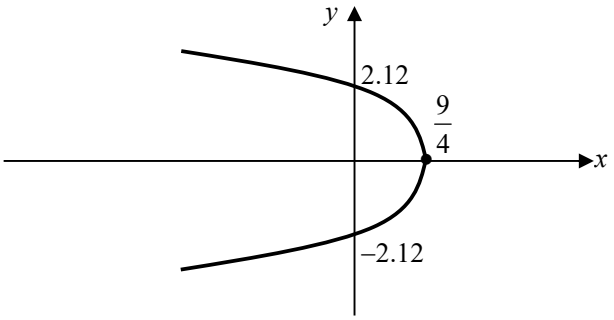
(ii) $(y-1)^2=2x+4 \Rightarrow (y-1)^2=2(x+2)$
Vertex of parabola is $(-2,1)$. Note that $a = 2 > 0$.
When $x = 0$, $y=1\pm\sqrt{4}=3$ or -1



(iii) $2y^2 + 4x - 9 = 0 \Rightarrow y^2 = -2x + \frac{9}{2} \Rightarrow y^2 = -2\left(x - \frac{9}{4}\right)$

Vertex of parabola is $\left(\frac{9}{4}, 0\right)$. Note that $a = -2 < 0$.

When $x = 0$, $y = \pm\sqrt{\frac{9}{2}} = \pm\frac{3}{\sqrt{2}} = \pm 2.12$ (3 sf).



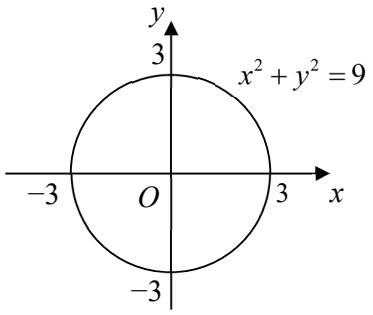
3.2 Circle

The equation of a circle with centre (0,0) and radius r units is given by

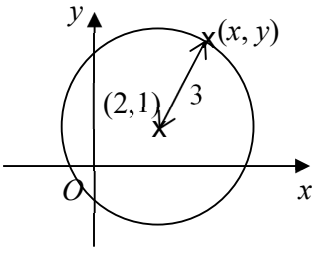
$$x^2 + y^2 = r^2, \quad \text{where } r > 0.$$

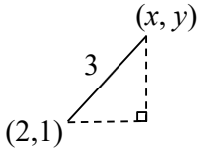
Consider the circle with equation $x^2 + y^2 = 9$.

This is a circle with centre $(0, 0)$ and radius 3 units.



Now, let us find out the equation of a circle with centre (2,1) and radius 3 units.





Using Pythagoras' Theorem, we know the equation of a circle with centre (2,1) and radius 3 units is $(x-2)^2+(y-1)^2=3^2=9$.

The standard form of the equation of a circle with centre (h,k) and radius r units is given by
$$(x-h)^2+(y-k)^2=r^2, \quad \text{where } r>0.$$

Note:

1. The coefficients of x^2 and y^2 are equal.
2. When sketching a circle,
   - the scale for both axes should be the same, and ***use a compass***,
   - label the centre (h, k) and radius r clearly,
   - check if the origin is in the circle, on the circle, or outside the circle. (How?)

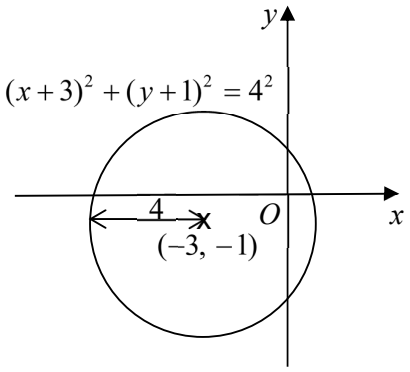
Example 13

Sketch, on separate diagrams, the circles given by the equations

(i) $(x+3)^2+(y+1)^2=16,$ **(ii)** $x^2+y^2+2x-4y=0$.

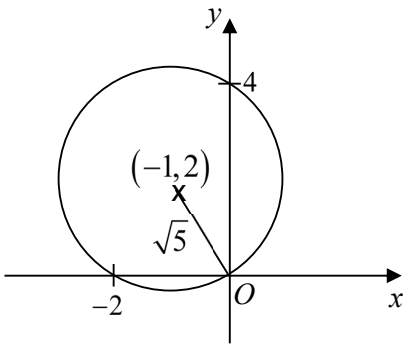
Solution

(i)

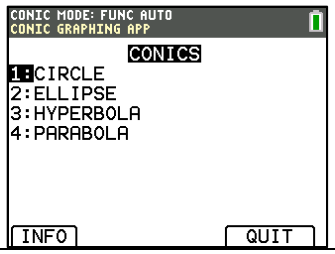


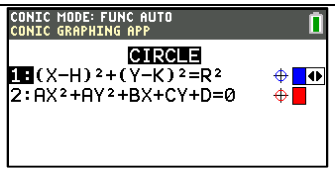
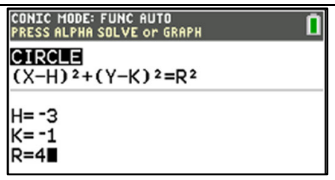
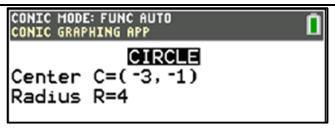
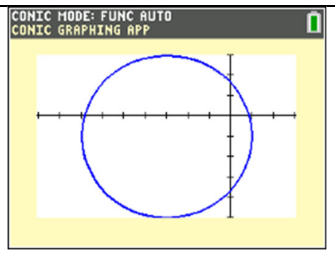
Circle with centre $(-3,-1)$
and radius 4 units.

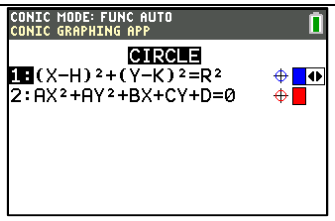
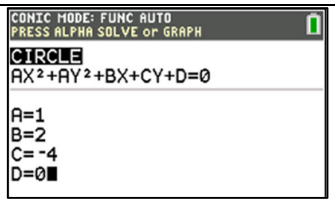
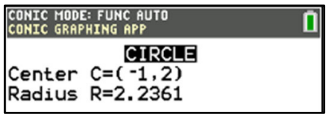
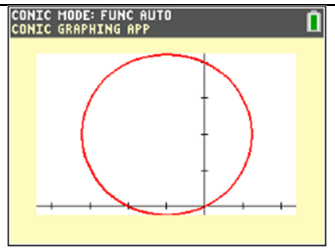
(ii)



Circle with centre $(-1,2)$
and radius $\sqrt{5}$ units.

GC Keystrokes for (i)	*Screenshots*
The **Conics Apps** in the GC can be used to sketch conics. In the **Conics App**, select **Circle**.	

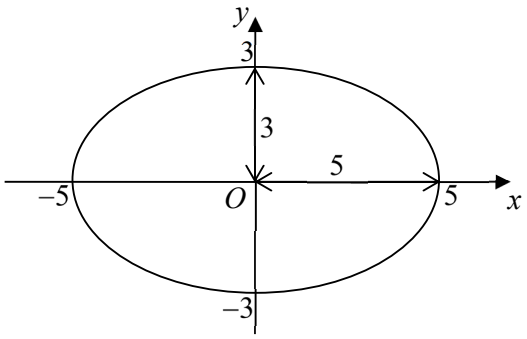
Sketch the circle given by the equation $(x+3)^2+(y+1)^2=4^2$ 1. Select the appropriate form, 1: $(x-h)^2+(y-k)^2=r^2$ in this case.	
2. Key in the values for the various constants.	
3. Press alpha enter to obtain the centre and radius.	
4. Press graph for a sketch of the circle.	

GC Keystrokes for (ii)	***Screenshots***
Sketch the circle given by the equation $x^2+y^2+2x-4y=0$ 1. Select the appropriate form, 2: $Ax^2+Ay^2+Bx+Cy+D=0$ in this case.	
2. Key in the values for the various constants. 3. Press alpha enter to obtain the centre and radius. **Limitation** : The GC can only give decimal values. So if the radius is $\sqrt{5}$, the GC will give 2.2361.	 
4. Press graph for a sketch of the circle.	

3.3 Ellipse

The equation of an ellipse centred at (0, 0) is given by $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, where $a>0, b>0$.

Consider the ellipse with equation $\frac{x^2}{5^2}+\frac{y^2}{3^2}=1$.



This is an ellipse with centre (0,0).

The graph is symmetrical about the *x*-axis and *y*-axis.

The semi-major axis is 5 units and

the -minor axis is 3 units.

Note:

1. When $a=b$, the equation becomes the equation of a circle of radius *a* and centered at the origin. Hence, a circle is a special ellipse.
2. When $a\neq b$, the x^2 and y^2 terms have the same sign but different coefficients.
3. When sketching an ellipse centered at the origin,
   - the scale for both axes should be the same,
   - note symmetry about the origin,
   - label the intercepts.

Example 14

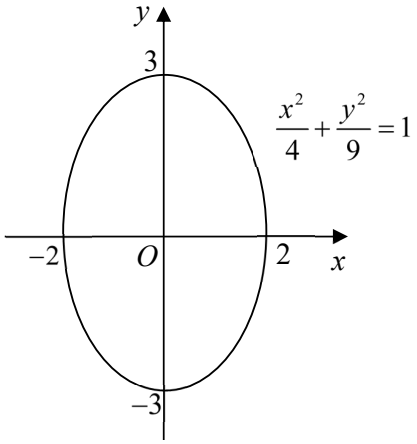
Sketch the ellipse with equation $9x^2+4y^2=36$.

Solution

$9x^2+4y^2=36$

Rearranging into standard form, we get

$\frac{x^2}{2^2}+\frac{y^2}{3^2}=1$



NOTE : You may refer to Appendix I (A) to see how we can use graphing calculator to help us sketch an ellipse.

3.4 Hyperbola

The equation of a hyperbola with centre (0,0) is given by

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1 \quad \text{or} \quad \frac{y^2}{a^2}-\frac{x^2}{b^2}=1$$

Consider the hyperbola with equation $\frac{x^2}{9}-\frac{y^2}{16}=1$.

This graph is symmetrical about the x–axis and y–axis.

It has 2 oblique asymptotes, $y=-\frac{4}{3}x$ and $y=\frac{4}{3}x$.

When $y=0$, $\frac{x^2}{9}=1 \Rightarrow x=\pm 3$

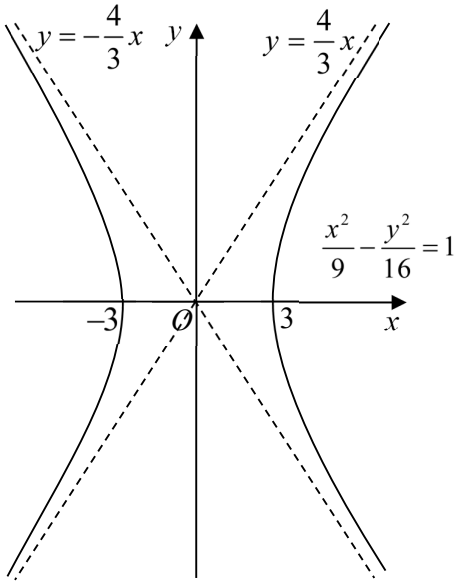
To find the asymptotes of a hyperbola :

Consider $\frac{x^2}{h^2}-\frac{y^2}{k^2}=1$. Rewriting, we obtain $\frac{y^2}{k^2}=\frac{x^2}{h^2}-1$.

Observe that as $x\to\pm\infty$, 1 becomes insignificant. Hence, $\frac{y^2}{k^2}\approx\frac{x^2}{h^2}$ for large values of x.

Consequently, we get $y=\pm\frac{k}{h}x$ as the oblique asymptotes.

Similarly for $\frac{y^2}{k^2}-\frac{x^2}{h^2}=1$.



<u>Not in syllabus: Formal proof of obtaining the asymptotes of a hyperbola</u>

Consider $\frac{x^2}{h^2}-\frac{y^2}{k^2}=1$.

Rewriting, we obtain $y=\pm\frac{k}{h}x\sqrt{1-\frac{h^2}{x^2}}$.

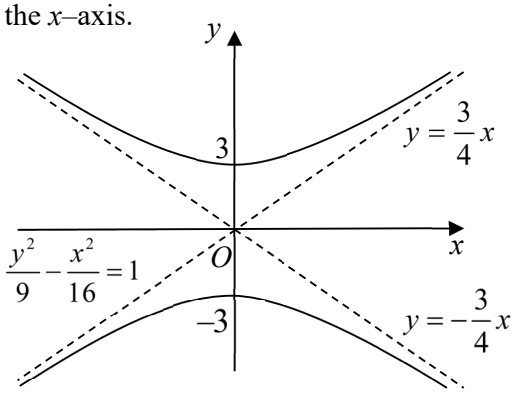
As $x\to\pm\infty$, $\frac{h^2}{x^2}\to 0$. Hence, $y\to\pm\frac{k}{h}x$.

Consequently, we get $y=\pm\frac{k}{h}x$ as the oblique asymptotes.

(Similarly for $\frac{y^2}{k^2}-\frac{x^2}{h^2}=1$.)

What about the graph of a hyperbola with equation $\frac{y^2}{9}-\frac{x^2}{16}=1$?

Using the same approach as above, we observe that this graph is symmetrical about the x–axis and y–axis. It also has 2 oblique asymptotes, $y=-\frac{3}{4}x$ and $y=\frac{3}{4}x$. However, in this case, the curve intersects the y–axis but not the x–axis.



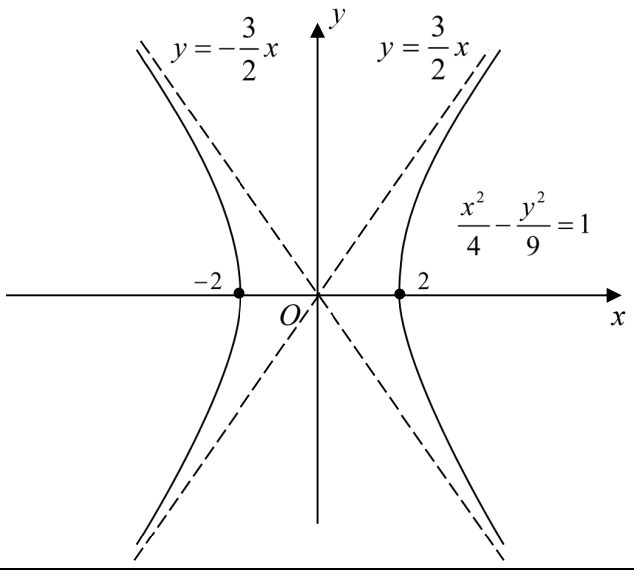
Note :

1. The x^2 and y^2 terms have opposite signs.
2. When drawing a hyperbola,
   - the scale for both axes should be the same
   - draw curve approaching the oblique asymptotes
   - label the vertices of the hyperbola
   - label the equations of the asymptotes.

Example 15

Sketch the hyperbola $\frac{x^2}{4}-\frac{y^2}{9}=1$, indicating clearly the equations of asymptotes.

Solution



NOTE : You may refer to Appendix I (B) to see how we can use graphing calculator to help us sketch a hyperbola.

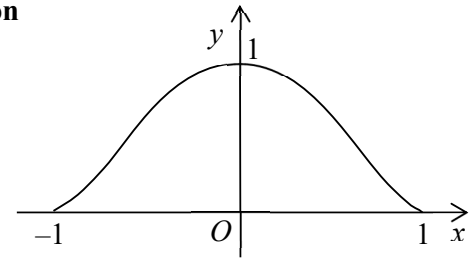
4 PARAMETRIC EQUATIONS AND THEIR GRAPHS

Recall that in Chapter 5A, we learnt how to find the gradient of a curve defined parametrically at any given point. In this section, we will learn how to sketch such parametric curves using the GC.

Example 16

Sketch the curve with parametric equations $x=\cos\theta, y=\sin^3\theta, 0\le\theta\le\pi$.

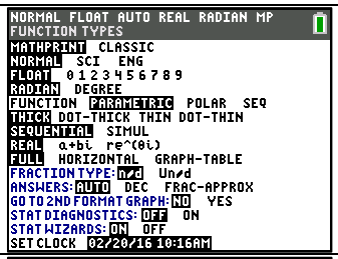
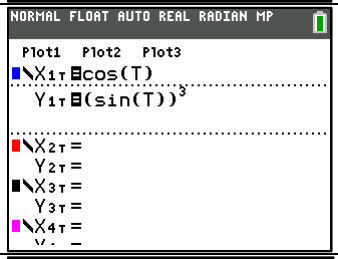
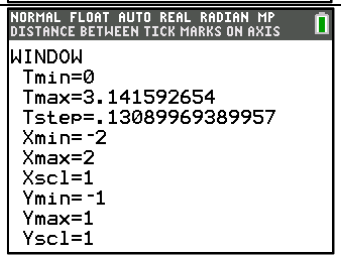
Solution

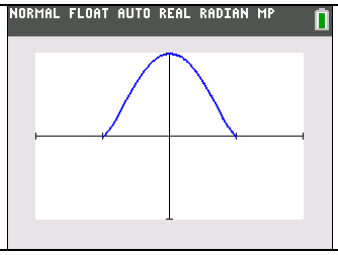


When $x=0,\ \theta=\dfrac{\pi}{2},\ y=1$

When $y=0,\ \theta=0$ or π

$x=1$ or -1

GC Keystrokes	*Screenshots*
1. Change the mode to **PARAMETRIC** (parametric mode).	
2. In the [y=] screen, enter the functions.	
3. Need to adjust window settings. **Limitation** : The default setting is $T_{min}=0, T_{max}=2\pi$. Remarks : T_{min} and T_{max} refers to the minimum value and maximum value respectively, taken by the parameter (θ in the case of Example 16) in the equations.	

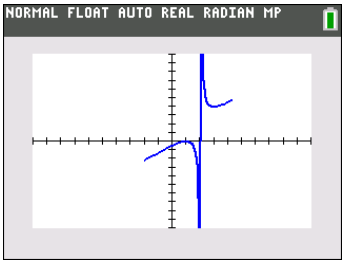
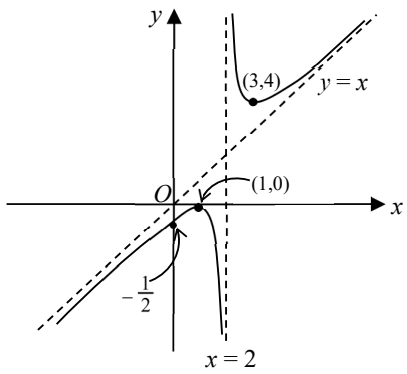
4. Press graph for a sketch of the curve.	

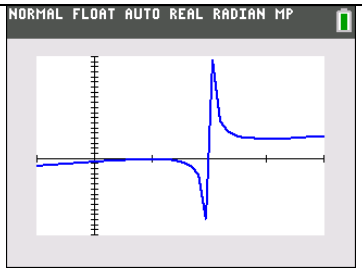
Note:
The GC gives the shape of the graph. In many cases, you may need to resize the window to capture the entire graph, or to change window settings to capture critical features of the graph.

However, **it does not draw asymptotes**!

To determine if a graph has any asymptotes, you need to examine its equation. Sometimes, a vertical line may appear like a vertical asymptote, but actually it is only connecting one point off the top of the screen with the next point off the bottom. The GC does not know if there is any undefined value between the values it is graphing.

To illustrate this point, we shall investigate a case by sketching the graph of
$x=t-2$ and $y=t-2+\dfrac{1}{t-4}$.

What you see	What it should be
$x=t-2$ and $y=t-2+\dfrac{1}{t-4}$  **<u>Remarks</u>** : There are actually 2 asymptotes ($y=x$ and $x=2$). The vertical line to the right of y-axis appears like a vertical asymptote, but actually it is only connecting one point off the top of the screen with the next point off the bottom. The GC does not know if there is any undefined value between the values it is graphing. This is made more obvious in another window setting as shown below :	$x=t-2$ and $y=t-2+\dfrac{1}{t-4}$ 



Example 17

A curve has parametric equations

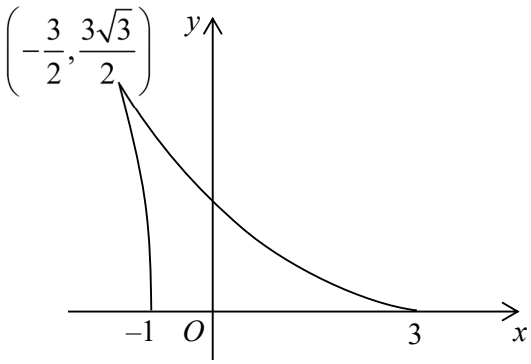
$$x = 2\cos\theta + \cos 2\theta, \quad y = 2\sin\theta - \sin 2\theta, \quad 0 \le \theta \le \pi .$$

(i) Sketch the curve.

(ii) Express x in the form $x = 2(\cos\theta + a)^2 + b$ where a and b are constants to be determined. Hence, find the least value of x and the value of θ when it occurs.

Solution

(i)



(ii)

$$\begin{aligned} x &= 2\cos\theta + \cos 2\theta \\ &= 2\cos\theta + 2\cos^2\theta - 1 \\ &= 2(\cos^2\theta + \cos\theta) - 1 \\ &= 2\left[\left(\cos\theta + \frac{1}{2}\right)^2 - \frac{1}{4}\right] - 1 \\ &= 2\left(\cos\theta + \frac{1}{2}\right)^2 - \frac{3}{2} \end{aligned}$$

where $a = \frac{1}{2}$ and $b = -\frac{3}{2}$.

Least value of x is $-\frac{3}{2}$ when $\theta = \frac{2\pi}{3}$

CONCLUSION

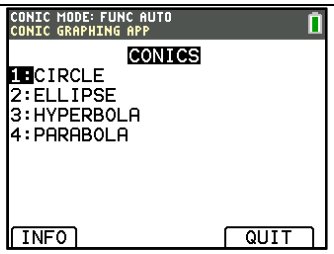
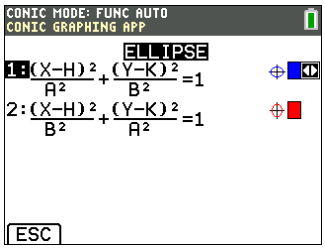
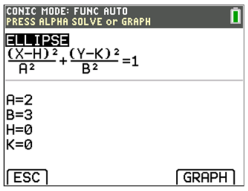
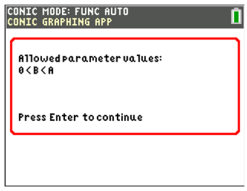
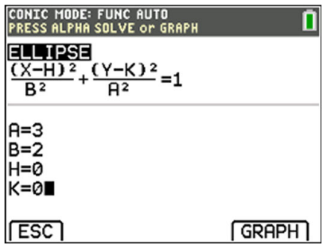
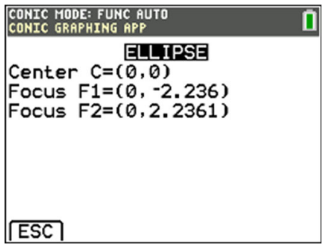
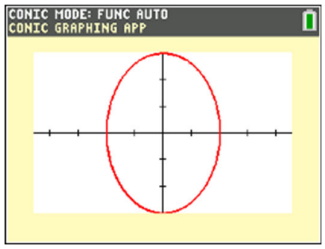
Graphs have many practical uses in everyday life. For example, the exponential curve is useful in modelling the rate of growth of the population of a species, or the rate of decay of a radioactive substance. The applications of conics are related to their reflective properties and to planetary motion. For example, the reflectors on automobile headlights and dish antennas are parabolic, while the orbits of planets are elliptic.

It is useful to recognise the equations of basic functions and understand the properties of their graphs. When sketching graphs, one should take note of characteristics such as symmetry, intersections with the axes, turning points and asymptotes.

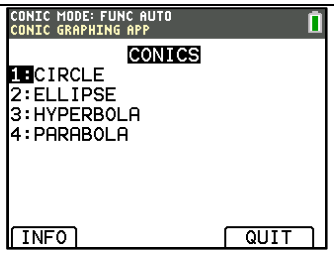
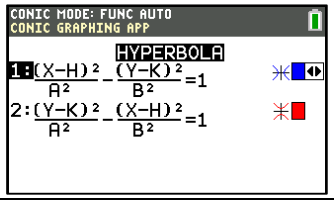
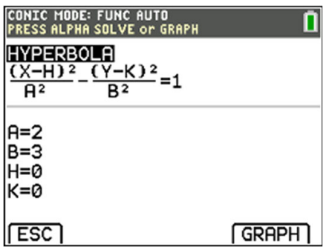
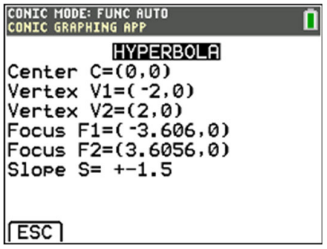
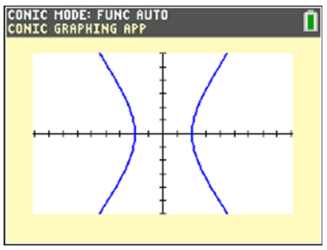
A graphic calculator is a useful tool in sketching graphs. However, it has certain limitations. For example, it cannot draw vertical asymptotes, and the graphs obtained are only an approximation. A GC also does not know what critical features of a graph to display – that is dependent on you.

Appendix I - Sketching ellipse and hyperbola using a graphing calculator

(A) Ellipse (Refer to Example 14)

GC Keystrokes	*Screenshots*
Sketch the ellipse with equation $$9x^2+4y^2=36.$$ Before using the GC, we need to rearrange the above equation into the form, $$\frac{x^2}{2^2}+\frac{y^2}{3^2}=1.$$ We also need to know the orientation of the ellipse and choose the correct form accordingly i.e. $A>B>0$. In the **Conics App**, select **Ellipse**. 1. Choose the appropriate form, 2 : $\frac{(x-h)^2}{b^2}+\frac{(y-k)^2}{a^2}=1$ in this case	 
2. Key in the values for the various constants (check that $A>B>0$, otherwise an Error, as shown below, will occur).  	
3. Press alpha enter to obtain the centre	
4. Press graph for a sketch of the ellipse	

(B) Hyperbola (Refer to Example 14)

GC Keystrokes	*Screenshots*
Sketch the hyperbola $\frac{x^2}{4}-\frac{y^2}{9}=1$. In the **Conics App**, select **Hyperbola**. 1. Choose the appropriate form, 1 : $\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$ in this case	 
2. Key in the values for the various constants	
3. Press alpha enter to obtain the centre, vertex and slope (gradient of asymptote)	
4. Press graph for a sketch of the hyperbola **Limitation :** The GC does not draw the asymptotes.	

SUMMARY